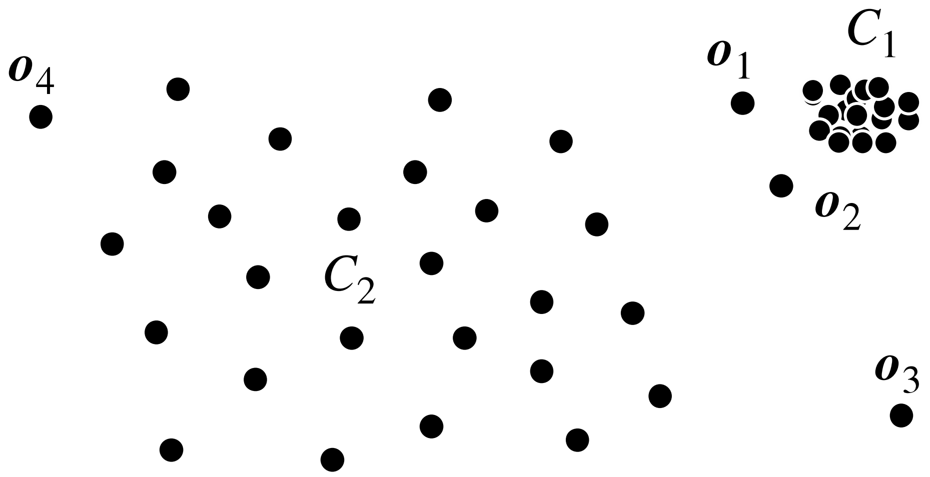
Density-Based Outlier Detection

- Local proximity-based outliers
- Challenge: how to measure the relative density of an object?



k-Distance

- The k-distance of o, denoted by $dist_k(o)$, is the distance, $dist(o, p)$, between o and another object, p ∈ D, such that
  - There are at least k objects $o' \in D\backslash\{o\}$ such that $dist(o, o') \le dist(o, p)$
  - There are at most k−1 objects $o'' \in D\backslash\{o\}$ such that $dist(o, o'') < dist(o, p)$
- The k-distance $dist_k(o)$ is the distance between o and its k-nearest neighbor
- The k-distance neighborhood of o contains all objects of which the distance to o is not greater than $dist_k(o)$, denoted by
$$N_k(o) = \{o' | o' \in D, dist(o, o') \le \ dist_k(o)\}$$
- $N_k(o)$ may contain more than k objects because multiple objects may each be the same distance away from o